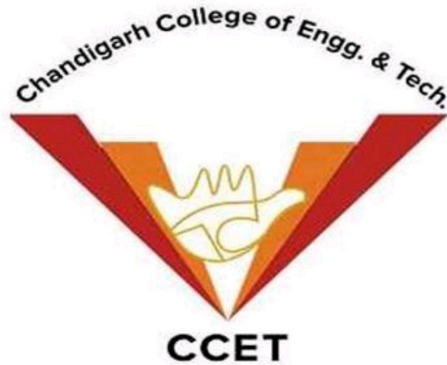




MAJOR PROJECT REPORT

ON

Transact Rite

- Your master companion for all financial transactions and note handling.

SUBMITTED TO –

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TRANSACT RITE

- Your master companion for all financial transactions and note handling.

Introduction:

TRANSACT RITE is the product of efforts put in by our prestigious group Tech Engineers.

Transact Rite was the perfect choice practically ,technically and user's accepted choice. Lets discuss the aim and objectives on these three grounds briefly,

TRANSACT RITE : a practical choice-

Looking at the advancements of finance and banking services on online portals and platforms it was very much a practical need to bring all these services at single and widely accessible Android platform.

Also with putting in our previous experience with minor project on ATM Simulator being desktop application , Transact Rite was the perfect step-up and choice that uses our knowledge and skillsets to their full potential.

TRANSACT RITE : a technical choice-

Now the very foundation of our team is built around the shared admiration and willingness to work with Java, the most versatile, robust and portable multipurpose language .

All of our team members possess knowledge as well as hands on experience with Java and OOPS concepts and this project in Android platform has provided us all an enriched and suitable advanced learning opportunity.

TRANSACT RITE : a user friendly choice-

Now , technology today is a tool that better serves the masses otherwise the masses need to better the tool !

Bring the financial services and processes at the user friendly and this widely popular choice of platform , Android is using today's technology is used at its finest ! Plus being the consumers of such services we also get to have a close look ,starting from designing going on to the final testing .

Features:

The proposed banking application aims to deliver a seamless and secure user experience, incorporating the following key features:

1. Authentication Processes-

The application prioritizes user security through robust authentication mechanisms. It includes:

- User Registration and Login: Secure onboarding of users with unique credentials through an easy-to-use interface. Ensures secure data storage using encryption.
- Two-Factor Authentication (2FA): Adds an extra layer of security by requiring both a password and a one-time password (OTP) sent via SMS or email.
- Password Recovery: Allows users to reset their password securely by verifying their identity through email or phone.
- Android-Based Authentication:
- Biometric Authentication:
 - Uses Android's FingerprintManager API and BiometricPrompt API for fingerprint and face recognition.
 - Ensures high-level security by storing biometric data locally on the device in a secure hardware-backed key store.
 - Provides a seamless login experience by eliminating the need to remember passwords.
- Google Sign-In Integration:
 - Simplifies authentication by allowing users to log in using their Google accounts.
 - Leverages the Google Identity Services API for secure and fast single sign-on (SSO).
 - Reduces barriers for new users by offering a familiar login process.

2. Transactions

The application offers smooth and efficient financial transaction capabilities, including:

- Funds Transfer: Allows users to send and receive money from bank to user or user to bank.
- Deposit and Withdrawal: Users can digitally deposit funds or request withdrawals through linked accounts or authorized agents.
- Balance Enquiry: Displays the current account balance in real-time.
- Mini Statement: Provides a quick view of recent transactions for easy account monitoring.
- Transaction History: Maintains a detailed log of all user transactions with filtering options.
- Scheduled Transactions: Enables users to schedule recurring payments or transfers.

3. Email Notification

This feature ensures transparency and keeps users informed about their account activities through:

- Transaction Alerts: Automated email notifications for completed, pending, or failed transactions.
- Promotional Updates: Updates on offers, promotions, and new features of the bank.
- Security Notifications: Alerts for login attempts or changes to account settings.

4. Currency Converter

To assist users dealing with international transactions, the app provides:

- Real-Time Conversion Rates: Displays up-to-date currency exchange rates.
- Multi-Currency Support: Allows users to convert and compare amounts in different currencies.
- User-Friendly Interface: Easy-to-use tools for quick calculations.

5. Online Payments

Facilitating digital banking, the application integrates secure online payment solutions, including:

- Bill Payments: Supports utility payments such as electricity, water, and internet bills.
- QR Code Payments: Allows users to scan QR codes for instant transactions.
- Payment Gateway Integration: Ensures compatibility with major online merchants and platforms.

Tools and technologies used:

Programming Languages:

- Java: The original programming language for Android apps. Known for its stability and extensive community support.
- Kotlin: A modern language preferred for Android development, offering concise syntax and full interoperability with Java.



Integrated Development Environment (IDE):

- Android Studio: The official development environment for creating Android applications, it's based on IntelliJ IDEA and provides numerous tools for building, testing, and debugging.



Build Tools:

- Gradle: An advanced build toolkit that manages dependencies, compiles code, and automates tasks in the development workflow.



Layout Design:

- XML (Extensible Markup Language): Used for designing UI layouts declaratively.
- Jetpack Compose: A modern toolkit for building native Android UIs using a declarative approach, which makes UI development more intuitive and efficient.



Android Jetpack Components:

- LiveData: Lifecycle-aware data holder class for updating the UI when data changes.
- ViewModel: Stores and manages UI-related data in a lifecycle-conscious way.
- Navigation: Handles in-app navigation, simplifying the implementation of user flows.
- Room: An abstraction over SQLite for database management and data persistence.
- Data Binding: Binds UI components in your layouts to data sources directly, reducing boilerplate code.

Debugging and Monitoring:

- Android Profiler: Provides real-time data to measure your app's characteristics.
- Firebase Crashlytics: A lightweight, real-time crash reporter that helps track, prioritize, and fix stability issues.



Source Control:

- Git: A distributed version-control system for tracking changes in source code.
- GitHub: A web-based platform for version control and collaboration.



Design and Prototyping Tools:

- Draw.io: An online tool for creating and collaborating on diagrams, integrating with cloud storage, and supporting offline use.

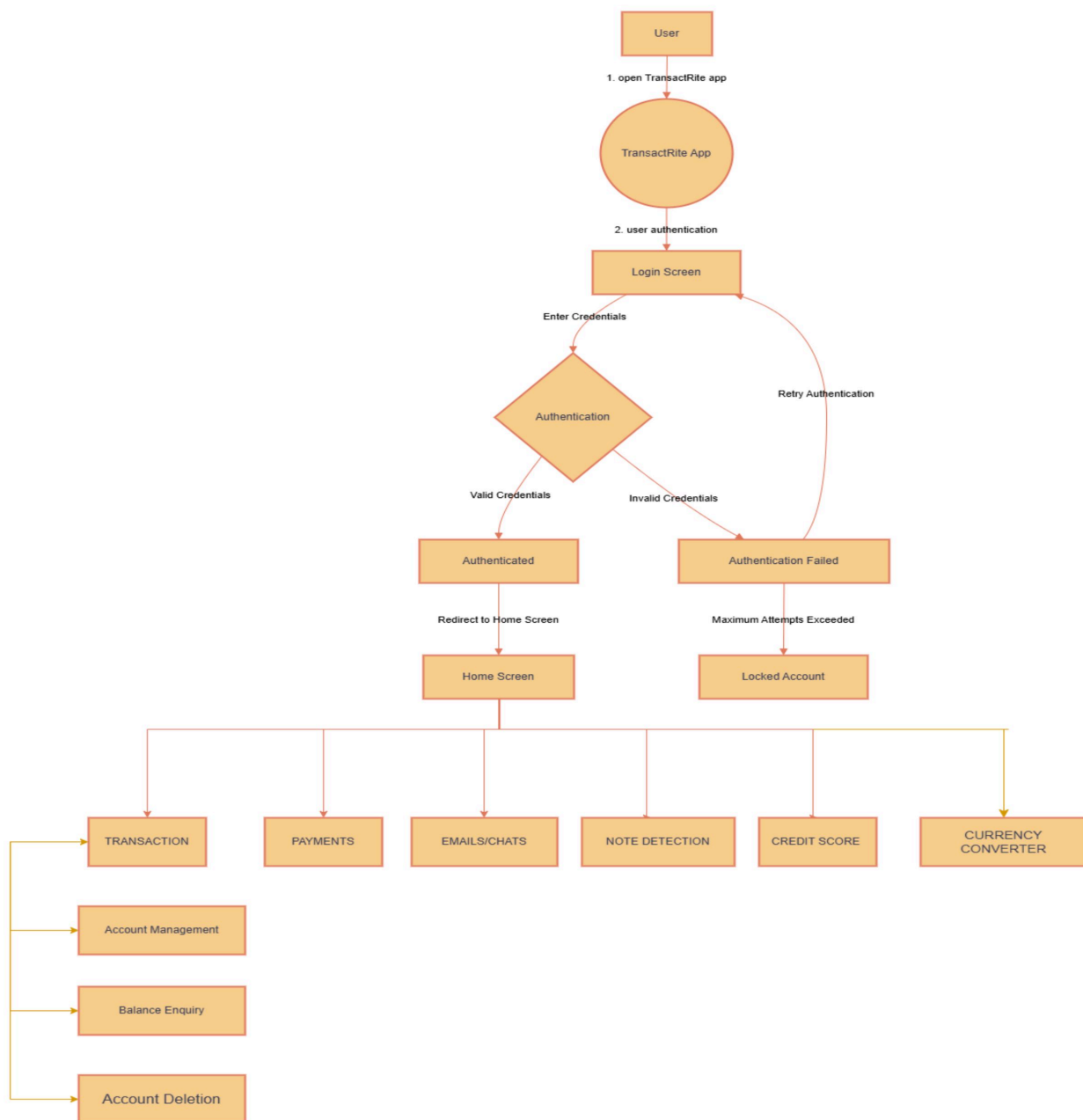
Libraries:

- Tensorflow: Google's open-source ML framework for building and training neural networks.
- Opencv: A library for real-time computer vision and image processing tasks.
- Mlkit: Mobile ML library by Google with easy APIs for on-device tasks like text recognition.

Dependencies:

- JUnit: Brings in JUnit for writing and running tests in Java projects.
- VectorDrawable: Supports scalable vector images, reducing APK size.
- CardView: Adds card-style UI elements with shadows and rounded corners.
- Legacy Support v4: Ensures backward compatibility for newer Android features.
- Biometric: Integrates fingerprint and face recognition for secure authentication.

Modules and Flow Diagram:



Hardware and Software Specifications:

1. Hardware Specifications:

i. Minimum Desktop Requirements-

- Operating System: Windows 7/8/10 (64-bit), macOS (10.10+), or Linux (GNOME or KDE desktop).
- RAM: Minimum of 4 GB RAM (8 GB RAM or more recommended).
- CPU: Intel Core i5 or higher (recommended), or equivalent AMD processor.
- Storage: Minimum of 2 GB disk space for Android Studio, plus at least 1 GB for Android SDK and emulator system images.
- Graphics: 1280 x 800 minimum screen resolution.

ii. Minimum Phone Requirements-

- Operating System: Android 8.0 (Oreo).
- RAM: 2 GB (minimum).
- CPU: Quad-core processor.
- Storage: At least 16 GB of internal storage.
- Screen Resolution: 720x1280 pixels (HD).

2. Software Specifications:

- Android Studio: Latest version - Android Studio Ladybug Feature Drop (2024.2.2).
- Java Development Kit (JDK): JDK 8 or above.
- Android SDK: SDK tools, platform tools, and the latest Android platform (ensure to include Android 8.0 (Oreo) SDK).
- Emulator: Install Android Emulator for testing.
- Gradle: Latest version of Gradle - v8.12.1.